

Quantum Simulation

From Hamiltonians to Adiabatic Computation

Ground states, Ising paths, and digital adiabatic methods

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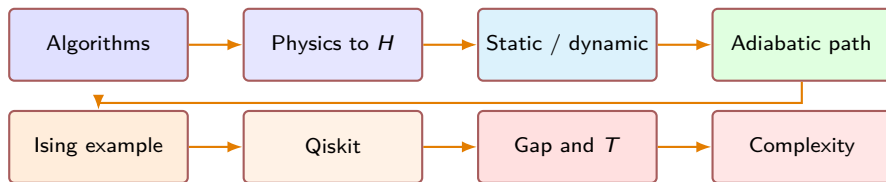
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- 10 Challenges and summary

- Quantum simulation: turn a physical model into a computational task
- Shared language: algorithms, Hamiltonians, and implementation methods
- Focus of this talk: adiabatic paths for ground-state problems, with a worked Ising example

Content outline

Physical model $\rightarrow H \rightarrow$ question $\rightarrow$ method $\rightarrow$ gap, runtime, and complexity.

Roadmap



Content outline

Deep dive on adiabatic ground-state computation; digital primitives appear as tools for the same path.

What Is an Algorithm?

- Finite procedure: input $\rightarrow$ instructions $\rightarrow$ output
- Independent of a particular machine realization
- “Quantum” changes the execution model, not this definition

Content outline

Operational definition; contrast with a device-specific program.

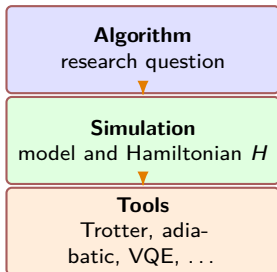
Quantum Algorithms at a Glance

- ① State preparation on qubits
- ② Unitary evolution: gate circuits or Hamiltonian control
- ③ Measurement and extraction of observables

Content outline

Three-part skeleton only; no gate-set or circuit-depth discussion yet.

Three Layers



Simulation is not a single gate.
It is a model-centered path:
question $\rightarrow H \rightarrow$ method.

Content outline

Layer 1 asks; layer 2 represents the physics; layer 3 implements $U(t)$, $H(s)$, or $f(H)$.

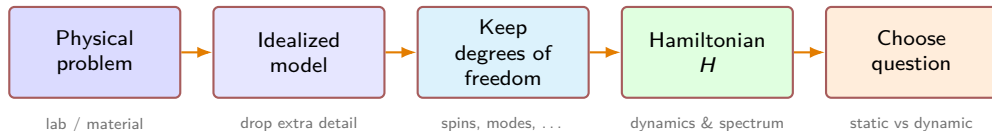
What Quantum Simulation Means

- Use a controllable quantum system to study another quantum model
- Digital, analog, or hybrid realizations
- Common starting point: fix H and the retained degrees of freedom

Content outline

Typical models: Hubbard, molecules, spin lattices, circuit modes.

From Physical Problem to Mathematical Model



Content outline

A model is a deliberate compression: what is kept versus discarded.

Representative Physical Problems

Physical setting	Typical model	Degrees of freedom
Magnets / spin glasses	Ising / Heisenberg	spins on a lattice
Correlated electrons	Hubbard	electrons + spin
Molecules	electronic structure	active orbitals
Cavities / circuits	oscillator / LC	flux and charge

Content outline

The same material can map to different models, depending on the question.

Classical and Quantum Roles of H

Classical

- Energy on phase space
- Generator of motion
- Minimization $\neq$ dynamics

Quantum

- Spectrum: $\hat{H}|E_n\rangle = E_n|E_n\rangle$
- Dynamics: $i\hbar \partial_t |\psi\rangle = \hat{H}|\psi\rangle$
- Same dual role continues

Content outline

Almost every simulation task is a function of one Hamiltonian: e^{-iHt} , ground state, or $H(s)$.

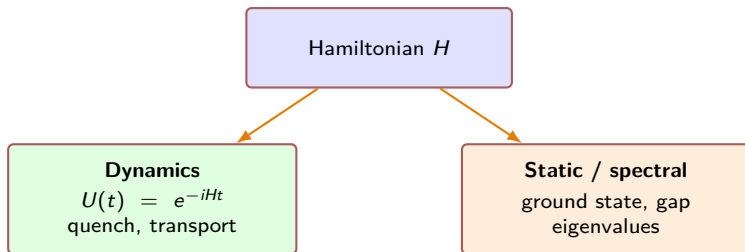
Why H Is Central in Simulation

- Fix the model and H first; then choose the question and the method
- Tools (Trotter, adiabatic, VQE) come after the representation is clear
- Choosing a method before fixing H is premature

Content outline

This talk keeps the Hamiltonian language primary.

Two Questions from One Hamiltonian



Content outline

The same H supports two research paths. This talk follows the static / ground-state branch.

Ground-State Problems: Why They Matter

- Lowest-energy configuration of a physical model (magnetism, molecules, lattices)
- Many optimization tasks map to Ising / QUBO ground states
- Spectral gap and low-lying excitations control both physics and runtime

Target for the rest of the talk

Prepare / identify the ground state of a problem Hamiltonian H_P .

Two Main Approaches

- ① **Time evolution:** build or approximate e^{-iHt} or $f(H)$
- ② **Adiabatic:** slow path $H_{\text{init}} \rightarrow H_P$, stay in the instantaneous ground state

Focus ahead

Adiabatic computation, a two-qubit Ising example, Qiskit implementation, then gap, runtime, and complexity.

Content outline

Trotter appears later as a digital tool for the same adiabatic path.

Adiabatic Idea in One Slide

- ① Start from an easy initial Hamiltonian H_{init} (H_B) with a known ground state
- ② Interpolate slowly to the problem Hamiltonian H_P
- ③ Sufficient gap and time $\Rightarrow$ ground state of H_P

Content outline

Encode the problem in a Hamiltonian path rather than enumerating configurations.

The Path $H(s)$

- Standard linear interpolation

$$H(s) = (1 - s) H_{\text{init}} + s H_P, \quad 0 \leq s \leq 1$$

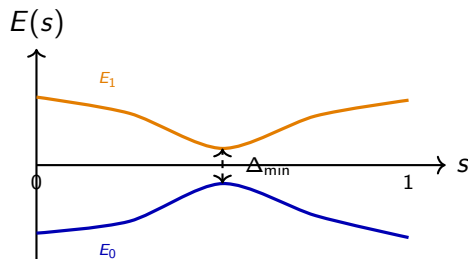
- Example schedule: $s(t) = t/T$
- s mixes the two Hamiltonians; it is not itself physical time

Content outline

$H(s)$ is linear in s ; the eigenvalues generally are not.

Instantaneous Gap and Runtime T

- $\Delta(s) = E_1(s) - E_0(s)$
- $\Delta_{\min} = \min_s \Delta(s)$ is the bottleneck
- Heuristic scale: $T \propto 1/\Delta_{\min}^2$



Content outline

Scaling estimate, not a universal theorem bound. Avoided crossing: levels approach, then separate.

Why Adiabatic for Ground States?

- Transparent physical picture: follow the ground state along a path
- Natural match to analog field schedules
- Digitally expressible via Trotterized paths (digital adiabatic)

Content outline

Less natural when the target is a precise e^{-iHt} or a general $f(H)$.

Two-Qubit Setup

- Make H_{init} , H_P , the gap, and T concrete
- System: two spin-1/2 particles (two qubits)
- Parameters: $h_x = 1$, $J = 1$, $h_z = 1/4$

Content outline

Small enough that the full spectrum can be seen numerically.

Initial Hamiltonian H_B

- $H_B = -h_x(X_1 + X_2)$ ($h_x > 0$)
- Easy ground state $|++\rangle$, prepared by Hadamards on $|00\rangle$
- Also called the driver / beginning Hamiltonian

Content outline

Transverse field; product eigenstates of X .

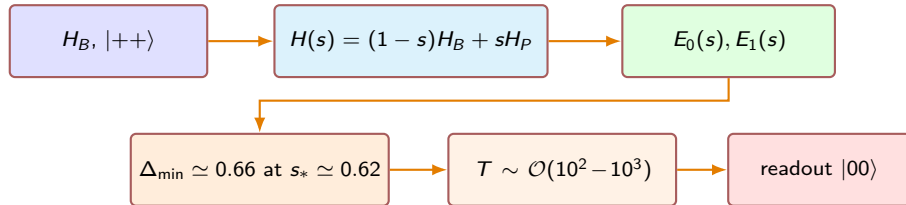
Problem Hamiltonian H_P

- $H_P = -JZ_1Z_2 - h_z(Z_1 + Z_2) \quad (J, h_z > 0)$
- Target ground state $|00\rangle$
- Ferromagnetic coupling plus a longitudinal field that lifts the degeneracy

Content outline

Optimization reading: the ground configuration is the solution.

Adiabatic Pipeline for the Ising Example

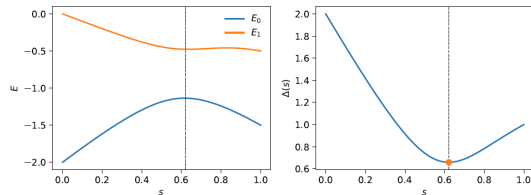


Content outline

Numbers from the two-qubit path; endpoint gaps alone do not locate the bottleneck.

Numerical Path Result

- Gap minimum near $s_* \simeq 0.62$
- $\Delta_{\min} \simeq 0.66$
- Heuristic with $\varepsilon \sim 10^{-2}$: $T_{\text{heur}} \sim 5 \times 10^2$
- For $n = 2$, $T \sim 40$ already yields near-unit success



Content outline

Distinguish a heuristic bound from a practical runtime checked by dynamics.

Role of Qiskit

- Shared implementation stack for digital circuits
- Turns the adiabatic path into a reproducible workflow
- Environment: `conda activate qcomputing`

Content outline

Close the loop from the Hamiltonian path to executable steps.

Code Steps (1): Build $H(s)$

- 1 Build H_B and H_P with `SparsePauliOp`
- 2 Form $H(s) = (1 - s)H_B + sH_P$
- 3 Scan s and extract $\Delta_{\min}$ and s_*

```
# Build H_B, H_P and the linear path H(s) (pedagogical excerpt)
from qiskit.quantum_info import SparsePauliOp

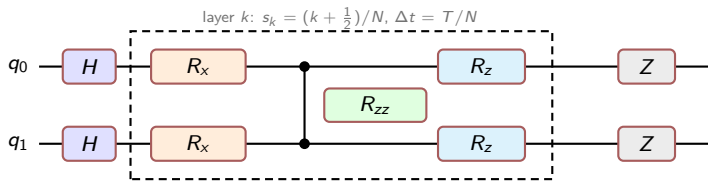
hx, J, hz = 1.0, 1.0, 0.25
HB = SparsePauliOp.from_list([("IX", -hx), ("XI", -hx)])
HP = SparsePauliOp.from_list([("ZZ", -J), ("IZ", -hz), ("ZI", -hz)])

def H_of_s(s: float) -> SparsePauliOp:
    return ((1.0 - s) * HB + s * HP).simplify()
```

Content outline

Scripts in `snippets/adi_ising_*.py` / notebook.

Code Steps (2): Trotter Layer and Readout



- $a = (1 - s)h_x\Delta t$, $b = sJ\Delta t$, $c = sh_z\Delta t$
- Prepare $|++\rangle$, apply layers, measure in the Z basis

Sample Execution Outcome

- With $T = 40$ and $N = 200$: $P(|00\rangle) \approx 0.9999$
- Aer shots concentrate on the bitstring 00
- The same run produces the path-gap plot

Content outline

Success probability is a check of the digital adiabatic schedule.

- **Instantaneous gap:** $\Delta(s) = E_1(s) - E_0(s)$
- **Minimum gap:** $\Delta_{\min} = \min_s \Delta(s)$ — the bottleneck
- **Avoided crossing:** levels approach then separate without crossing
- Full gap versus the gap inside a reachable symmetry sector

Content outline

The smallest spectral distance alone is not a sufficient criterion.

Three Meanings of Required Time

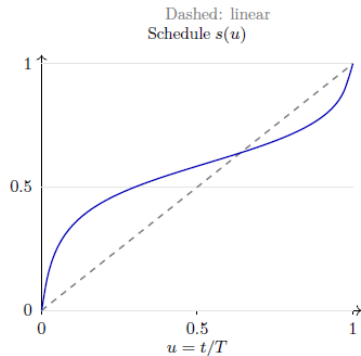
- ① Heuristic / order-of-magnitude estimate ($T \sim 1/\Delta_{\min}^2$)
- ② Sufficient bound under explicit assumptions
- ③ Practical time from dynamics and a success threshold

Content outline

These three numbers usually disagree.

Scheduling: Linear or Nonlinear

- Linear: $s(t) = t/T$
- Nonlinear: slow down near the bottleneck
- Same path $H(s)$, different speed through s
- Spend more time where $\Delta(s)$ is smallest



$$n \longrightarrow H_n(s) \longrightarrow \Delta_{\min}(n) \longrightarrow T(n) \longrightarrow \text{complexity}$$

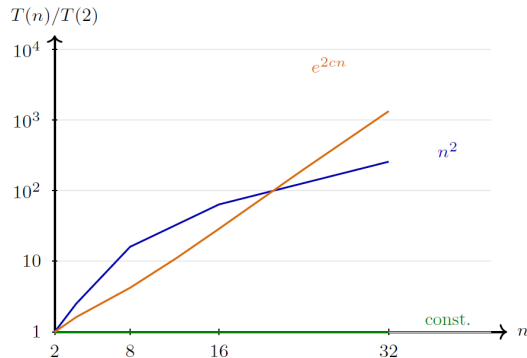
- Qubit count alone is not a measure of difficulty
- The family $H_n(s)$ decides how the gap closes

Content outline

Two problems with the same n can have completely different adiabatic costs.

Polynomial vs. Exponential Gap Closing

- Polynomial: $\Delta_{\min}(n) \sim n^{-\alpha} \Rightarrow T \sim n^{2\alpha}$
- Exponential: $\Delta_{\min}(n) \sim e^{-cn} \Rightarrow T \sim e^{2cn}$
- Polynomial closing is the scalable regime for adiabatic methods



Content outline

Relative runtime $T(n)/T(2)$: constant, n^2 , and e^{2cn} regimes.

Three Typical Regimes

Family	Gap	Typical T
Only initial H_B	$\Delta = 2h_x$ (indep. of n)	T independent of n
Critical TFIM (open)	$\Delta \sim 1/n$	$T \sim n^2$
Tunneling / ordered phase	$\Delta \sim e^{-cn}$	$T \sim e^{2cn}$

Content outline

Times are scaling estimates $T \sim \hbar/\Delta_{\min}^2$, not theorem bounds.

Three Complexity Layers

- ① Hardness of the **ground-state problem**
- ② Hardness of a **particular adiabatic path**
- ③ Cost of a **digital implementation** of that path

Message

Keep these three layers separate when assessing feasibility.

What Limits Adiabatic Simulation in Practice

- Gap closing with system size (polynomial vs. exponential)
- Control / scheduling near the bottleneck
- Noise and symmetry breaking along analog or digital paths
- Trotterization error and circuit depth in digital implementations

Content outline

Related routes for the same H_P : QAOA, VQE — different method, same model.

Key Takeaways

- 1 Simulation starts by fixing H and the degrees of freedom
- 2 Static and dynamic questions are distinct for the same H
- 3 Adiabatic computation is a path $H(s)$; the gap sets the time
- 4 Complexity follows $\Delta_{\min}(n)$ and the problem class, not n alone
- 5 Worked Ising + Qiskit make the gap and runtime concrete

- Adaptive schedules that slow down near a small gap
- Scaling studies beyond $n = 2$ (TFIM families)
- Comparison with variational routes such as QAOA and VQE

Content outline

Related notes: `QS&QA_Foundation/QS&QA_Foundation.tex`

Questions